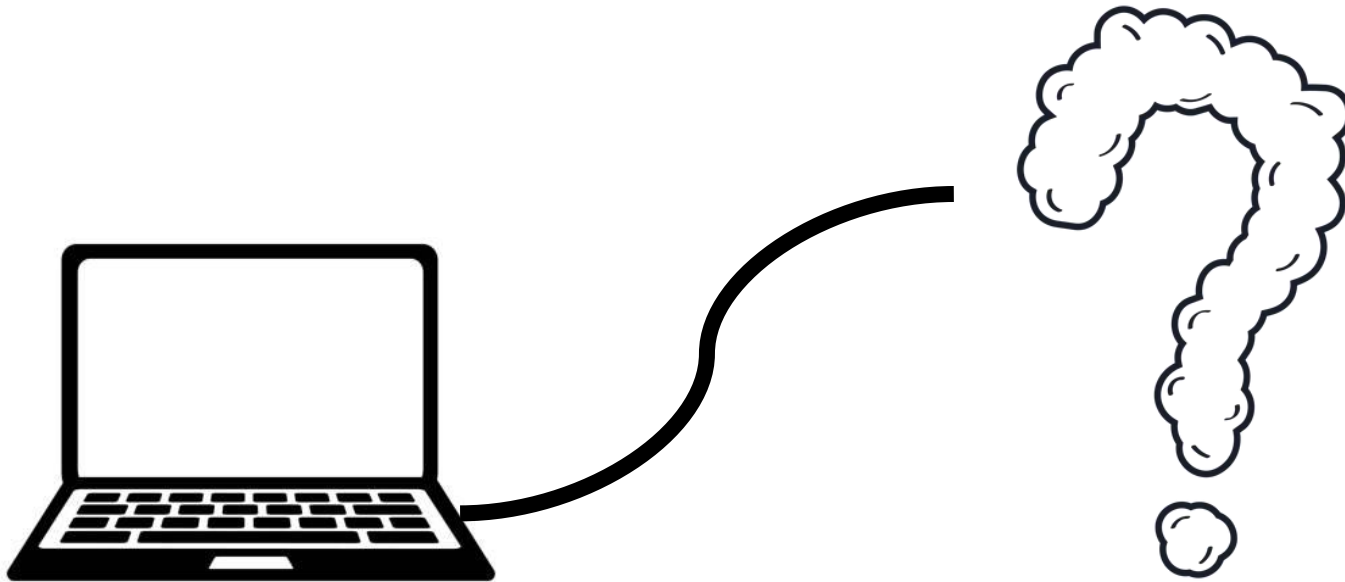


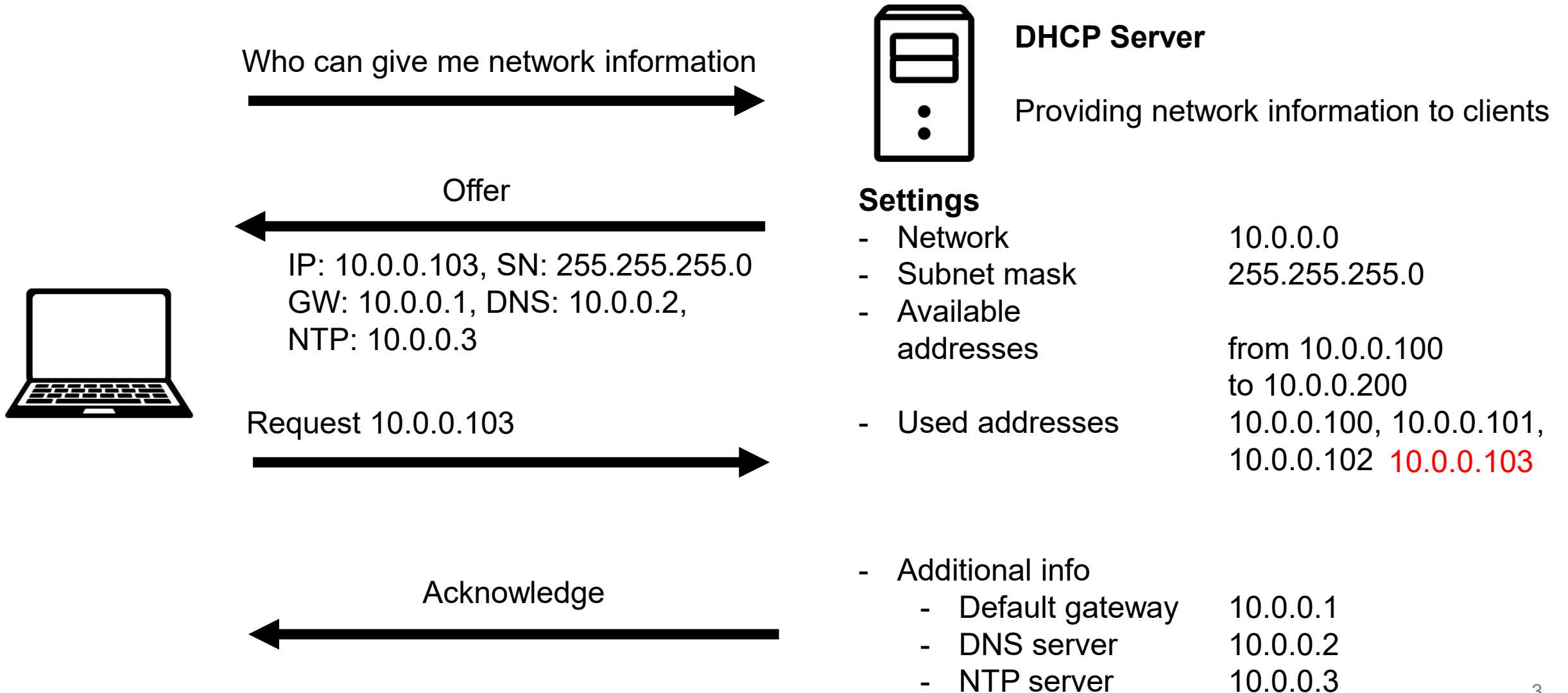
Network Fundamentals

The human network

Overview of common network protocols



Getting network information



Getting the time

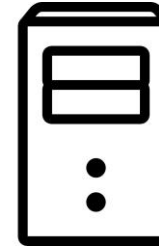


IP: 10.0.0.103,
SN: 255.255.255.0
Network: 10.0.0
Default gateway: 10.0.0.1
DNS: 10.0.0.2
NTP: 10.0.0.3

To 10.0.0.3, do you have the time



The time is



NTP Server

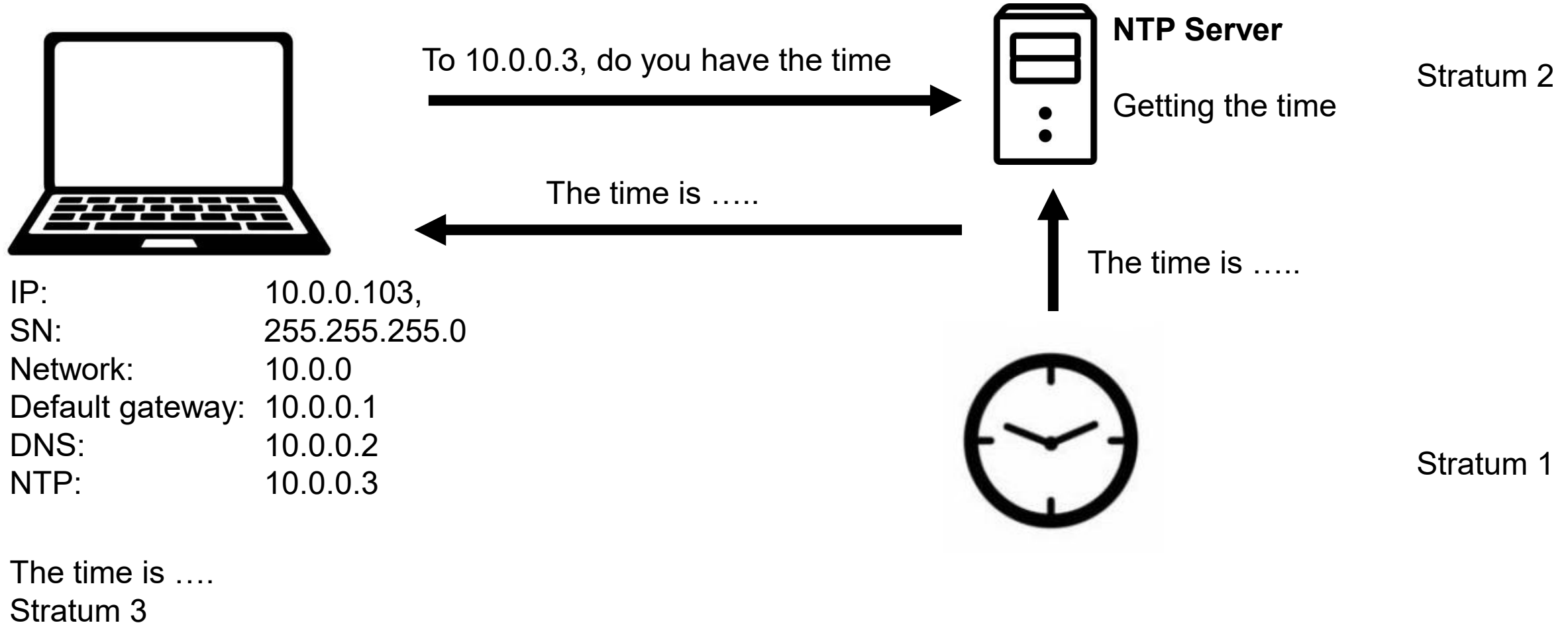
Getting the time



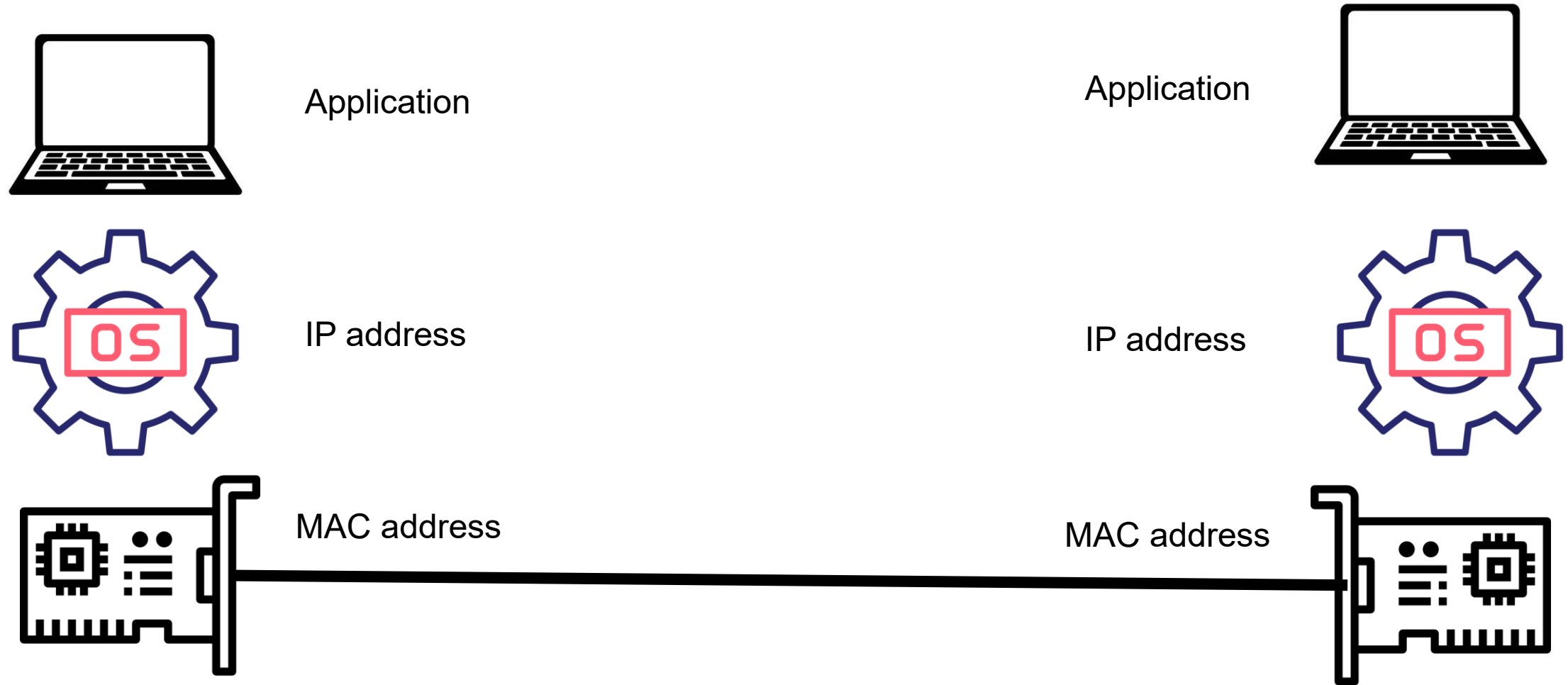
The time is



Getting the time



Logical and physical addresses



Logical and physical addresses

IP address

- Logical address
- Assigned on OS
- Can change
- Assigned depending on network
- Controls how devices communicate on a global scale

MAC Address

- Physical address
- Assigned on NIC
- Cannot change
- Hardcoded by manufacturer
- Identifies devices on a local network

Getting the physical address: ARP



Who is 10.0.0.120?

IP: 10.0.0.103,
SN: 255.255.255.0
Network: 10.0.0
Default gateway: 10.0.0.1
DNS: 10.0.0.2
NTP: 10.0.0.3

MAC: AA:AA:AA:11:11:11

Getting the physical address: ARP



IP: 10.0.0.103,
SN: 255.255.255.0
Network: 10.0.0
Default gateway: 10.0.0.1
DNS: 10.0.0.2
NTP: 10.0.0.3

MAC: AA:AA:AA:11:11:11

Who is 10.0.0.101?



IP: 10.0.0.101,
SN: 255.255.255.0
Network: 10.0.0
Default gateway: 10.0.0.1
DNS: 10.0.0.2
NTP: 10.0.0.3

MAC: BB:BB:BB:22:22:22

I am 10.0.0.101,
my mac is BB:BB:BB:22:22:22



ARP Table

10.0.0.101 → BB:BB:BB:22:22:22

Testing network connection: Ping



Send **to** 10.0.0.101, BB:BB:BB:22:22:22
Send **from** 10.0.0.103, AA:AA:AA:11:11:11
→
Are you there?



IP: 10.0.0.103,
SN: 255.255.255.0
Network: 10.0.0
Default gateway: 10.0.0.1
DNS: 10.0.0.2
NTP: 10.0.0.3

MAC: AA:AA:AA:11:11:11

←
Send **to** 10.0.0.103, AA:AA:AA:11:11:11
Send **from** 10.0.0.101, BB:BB:BB:22:22:22
Yes, I am

IP: 10.0.0.101,
SN: 255.255.255.0
Network: 10.0.0
Default gateway: 10.0.0.1
DNS: 10.0.0.2
NTP: 10.0.0.3

MAC: BB:BB:BB:22:22:22

ARP Table
10.0.0.101 → BB:BB:BB:22:22:22

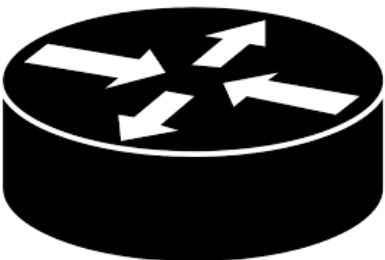
ARP Table
10.0.0.103 → AA:AA:AA:11:11:11

Testing network connection to remote device



ping 172.16.0.100

Not in my own network, send it to default gateway



IP: 10.0.0.103,
SN: 255.255.255.0
Network: 10.0.0
Default gateway: 10.0.0.1
DNS: 10.0.0.2
NTP: 10.0.0.3

MAC: AA:AA:AA:11:11:11

IP: 10.0.0.1,
SN: 255.255.255.0

MAC: CC:CC:CC:33:33:33

Who is 10.0.0.1?

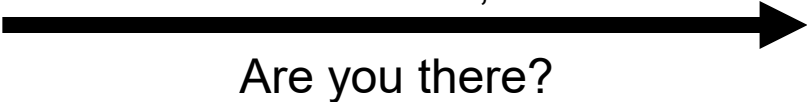


I am 10.0.0.1,
my mac is CC:CC:CC:33:33:33

ARP Table
10.0.0.101 → BB:BB:BB:22:22:22
10.0.0.1 → CC:CC:CC:33:33:33

Send to 172.16.0.100, CC:CC:CC:33:33:33

Send from 10.0.0.103, AA:AA:AA:11:11:11



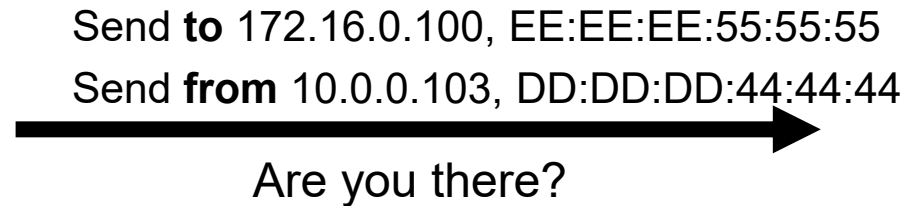
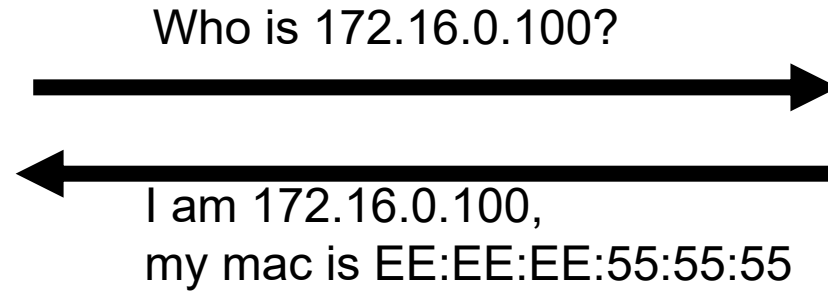
MAC address on a remote network



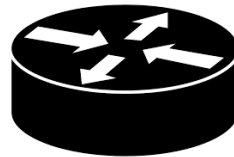
Send **to** 172.16.0.100, CC:CC:CC:33:33:33

Send **from** 10.0.0.103, AA:AA:AA:11:11:11

Are you there?



Left network connection



Right network connection

IP: 10.0.0.1

MAC: CC:CC:CC:33:33:33

IP: 172.16.0.1

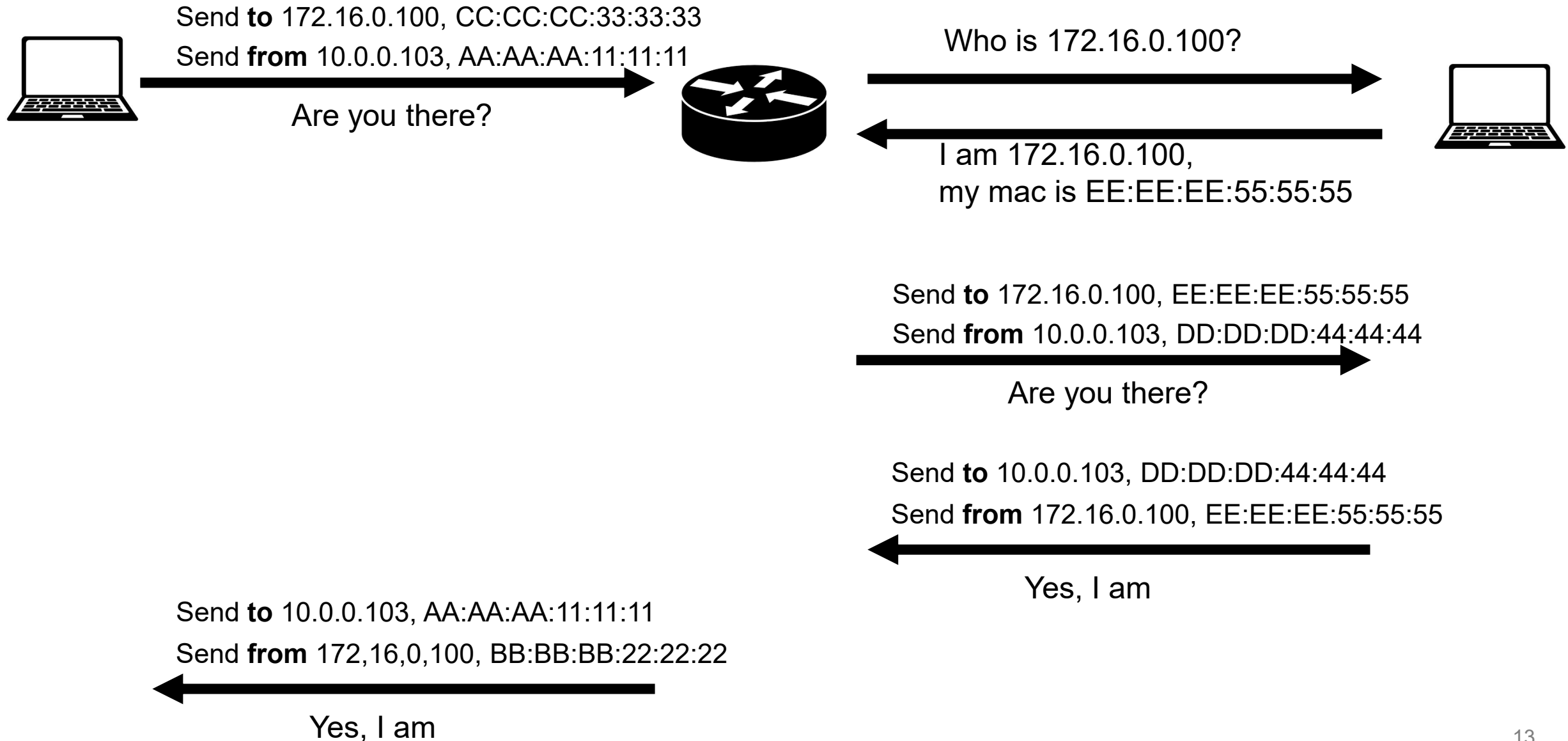
MAC: DD:DD:DD:44:44:44

ARP Table

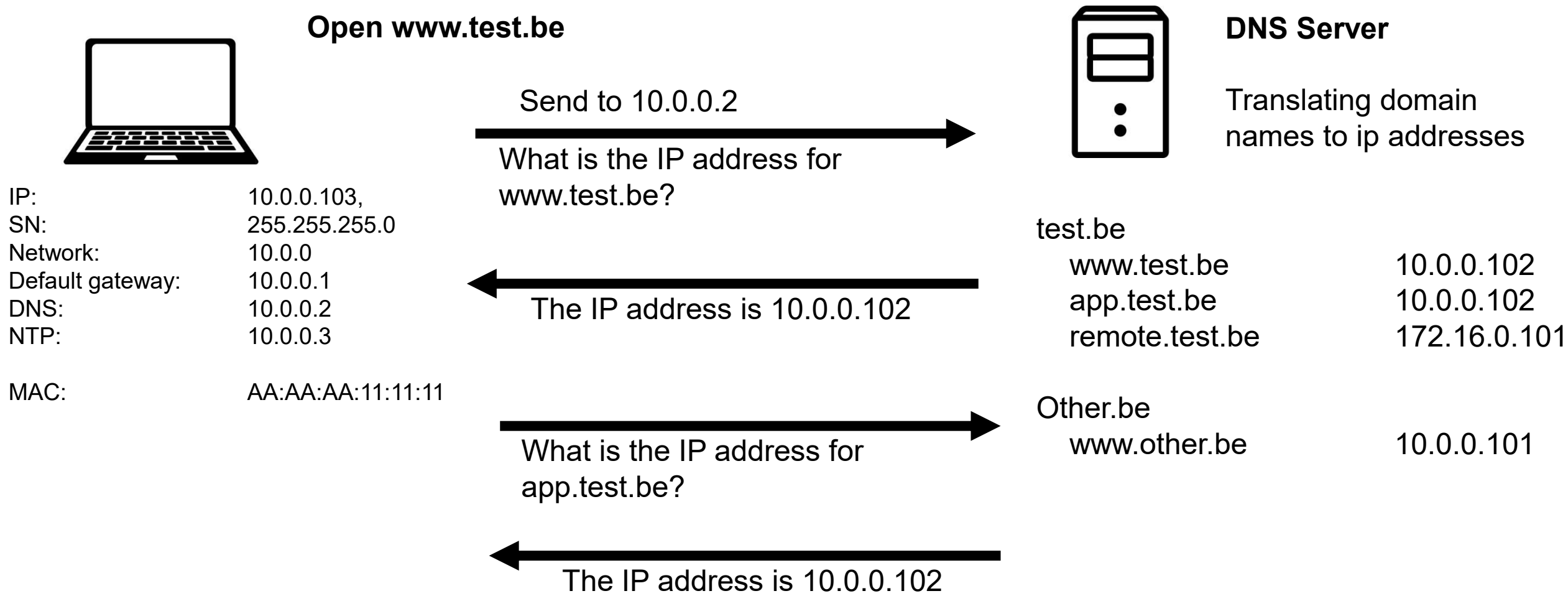
10.0.0.103 → AA:AA:AA:11:11:11

172.16.0.100 → CC:CC:CC:33:33:33

MAC address on a remote network



Resolving a domain name



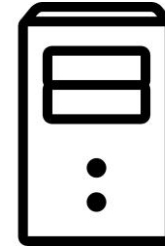
Reaching a domain name



Open `www.thomas.be`



What is the IP address for
`www.thomas.be`?



DNS Server

`test.be`

`www.test.be`

`10.0.0.102`

`app.test.be`

`10.0.0.102`

`remote.test.be`

`172.16.0.101`

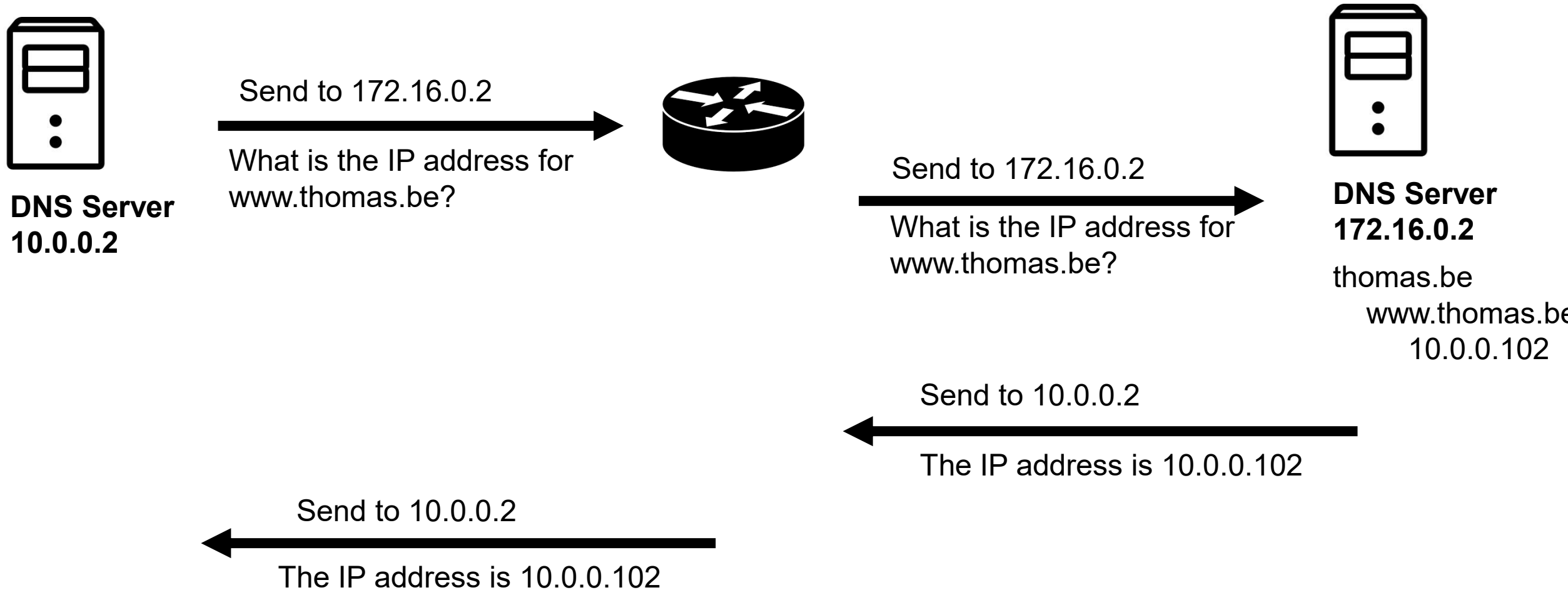
`Other.be`

`www.other.be`

`10.0.0.101`

Forwarder: `172.16.0.2`

Reaching a domain name

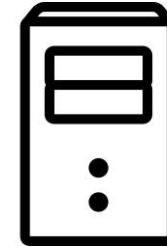


Reaching a domain name



Open `www.thomas.be`

What is the IP address for
`www.thomas.be`?



DNS Server

test.be	
www.test.be	10.0.0.102
app.test.be	10.0.0.102
remote.test.be	172.16.0.101

Other.be	
www.other.be	10.0.0.101

Forwarder: 172.16.0.102



The IP address is 10.0.0.102

Opening a webpage: http

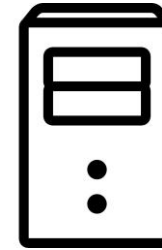


Open
`http://www.test.be/test.html`

Send to 10.0.0.102

Get:
`www.test.be/hello.html`

Ok:
`<h1>Hello World!</h1>`



Webserver

Serving webpages

hello.html



test.html



Misconfiguration: wrong DNS

Open `www.test.be`



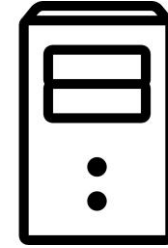
IP: 10.0.0.103,
SN: 255.255.255.0
Network: 10.0.0
Default gateway: 10.0.0.1
DNS: 10.0.0.3
NTP: 10.0.0.3

MAC: AA:AA:AA:11:11:11

Who is 10.0.0.3?

I am 10.0.0.2,
my mac is AC:AC:AC:13:13:13

What is the IP address for
`www.test.be`?



NTP Server

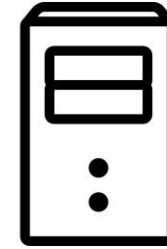


Misconfiguration: wrong dns config

Open <http://www.test.be/home.html>



What is the IP address for
www.test.be?



DNS Server

test.be

www.test.be

10.0.0.1

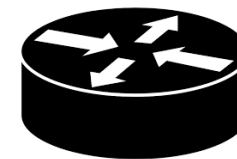


The IP address is 10.0.0.1



Send to 10.0.0.1

Get: www.test.be/home.html



Problem: non existing webpage

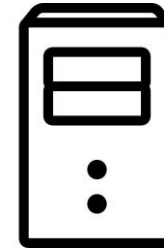


Open `http://www.test.be/wrongpage.html`

Send to 10.0.0.102

Get:
`www.test.be/error.html`

Error: Page not found



Webserver

Serving webpages

`hello.html`



`test.html`



Misconfiguration: wrong default gateway

